T2D Glucose Digital Twin

Predictive Metabolic Monitoring via Physics-Residual Temporal Fusion Transformers

Happiest Health Digital Twin Challenge — Final Submission

Target Use Case: Early Warning & Continuous Glucose Forecasting for Prediabetes and Type 2 Diabetes.

Team: T2D Digital Twin Team (Akshaya R — BGS College of Engineering and Technology; Nikhil VS — East West Institute of Technology)

Target Platform: Web-based Clinical Workstation & Replay Dashboard.

Release Commit: 67431af93f4e32d68c9df8c57e7ad24da9419de8

Investigational Research Prototype Notice: This software is an engineering research prototype designed to explore physiological modeling and deep learning integration on synthetic cohorts. It has not been clinically evaluated, certified as a medical device, or tested for autonomous therapeutic decisions.

RESEARCH PROTOTYPE

From a Reactive Reading to a Predictive Trajectory

The Clinical Problem and Monitoring Hypothesis

The Current Standard: Reactive Point Alarms

- Traditional Continuous Glucose Monitors (CGMs) alert after glucose breaches static thresholds (e.g., >180 mg/dL).
- Post-prandial glucose excursions occur rapidly (30–60 min rise time); threshold alarms leave minimal intervention window.
- Lack of physiological context leads to alert fatigue and high false alarm rates.

Our Approach: Continuous Forward Visibility

- Forecast future glucose trajectories across a **120-minute horizon** in 5-minute discretizations.
- Quantify predictive uncertainty via **calibrated quantile intervals** (p05 to p95).
- Estimate cumulative spike hazard and adverse event probability in real-time.
- Decompose contributing factors into mechanistic and behavioral drivers.

Clinical Boundary: The clinical hypothesis is that forward visibility aids timely dietary review; measured patient health outcomes or HbA1c reductions have not yet been evaluated in clinical trials.

Two Information Streams, One Patient Context

Observable Static EHR and Dynamic Telemetry Ingestion

Stream 1: Static EHR Patient Context

Captured at baseline / onboarding via validated schema:

- Demographics: Age, sex, BMI, weight.
- Metabolic Markers: Baseline HbA1c, eGFR, basal fasting glucose.
- Clinical History: Diabetes duration, diagnostic status (Healthy, Prediabetes, T2D).
- Medication Context: Metformin, SGLT2i, GLP-1 RA, DPP-4i, insulin.

Stream 2: Dynamic Telemetry & Wearable Streams

Ingested continuously on a synchronized 5-minute temporal grid:

- Continuous Glucose Monitoring (CGM) readings (mg/dL).
- Cardiorespiratory Wearable Signals: Heart Rate (HR), Heart Rate Variability (HRV).
- Physical Activity & Sleep: Step counts, sleep stage indicators.
- Logged Dietary Events: Carbohydrate intake timestamps and meal size.

Strict Observable-Input Boundary (Contract S3)

The deployed model strictly consumes observable clinical and wearable features. Privileged simulation latents (unannounced meal sizes, true plasma insulin) are isolated to the simulator environment and prohibited from entering model inputs.

VERIFIED INPUT CONTRACT

Hybrid Architecture: Physics Prior + Deep Residuals

Unscented Kalman Filter S3 + Temporal Fusion Transformer

1. Mechanistic UKF Prior

Two-compartment Bergman minimal glucose-insulin model with gut absorption dynamics:

- Tracks unobserved physiological states (insulin sensitivity, carbs-on-board).
- Provides deterministic physiological rollouts over 120 minutes.

2. PR-TFT Neural Architecture

Physics-Residual Temporal Fusion Transformer:

- Variable Selection Networks (VSN) for static and time-varying inputs.
- LSTM encoder/decoder capturing temporal dependencies.
- Causal multi-head self-attention for long-range interactions.

3. Residual Combination

The neural network learns the *residual error* over the physics prior:

- Final forecast = UKF physiological prior + PR-TFT learned quantile residuals.
- Predicts 7 quantiles (p05, p10, p25, p50, p75, p90, p95).
- Separate hazard head predicts spike probability.

Architectural Finding: Arm D (PR-TFT + UKF S3) enforces physiological plausibility during data gaps while allowing deep learning to capture complex non-linear habits and wearable interactions.

What the Model Returns: Trajectories & Risk

Standardized ForecastResponse v1 API Schema

Multi-Horizon Quantile Trajectory

- **Temporal Horizon:** 120 minutes forward from current as_of timestamp.
- **Discretization:** 24 discrete 5-minute timesteps.
- **Probabilistic Quantiles:** p05, p10, p25, p50 (median), p75, p90, p95.
- **Conformal Calibration:** Saved conformal interval widening fits the empirical 80% (p10–p90) confidence band with isotonic hazard calibration.

Adverse Event Risk Estimation

- **Spike Event Definition:** Glucose rising ≥ 50 mg/dL within 60 min and exceeding 180 mg/dL for ≥ 15 min.
- **Cumulative Risk Curve:** Onset probability evaluated across 5-min to 120-min offsets.
- **Risk Tiers:** Low (<35%), Watch (35–60%), High/Warning (≥60%).
- **Stateful Alert Policy:** Hysteresis filtering and 120-min suppression prevent alert flapping.

Verified Contract: All output packets strictly validate against Zod and Pydantic ForecastResponse schemas before reaching client presentation layers.

Clinician-Facing Workstation & Telemetry Stream

Interactive Next.js Dashboard and Bounded SSE Connection

Cohort Navigation

/cohort directory allows searching and filtering patients by clinical status (Healthy, Prediabetes, T2D) and split partition.

Patient Workspace

/patients/[id] displays real-time glucose ribbon, current trend arrow, physiological twin parameters, and risk waterfall.

Live Telemetry Stream

Server-Sent Events (SSE) deliver real-time readings. Bounded backoff recovery handles transient network disruptions automatically.

Post-Merge Verified SSE Reliability Pipeline

Following PR #3 merge (commit 68f8cce8), the telemetry pipeline enforces:

- **Chunk-Invariant Boundary Parsing:** Earliest-delimiter selection handles mixed CRLF/LF packet framing across all chunk partitionings.
- **Envelope-Level Validation:** Cross-subject packets are filtered on the SSE envelope without contaminating active patient state.
- **Companion Event Isolation:** Alerts and data quality changes update respective fields without wiping or resetting forecast ribbons.
- **Fail-Closed Security:** Malformed forecasts pass through to client validation to raise terminal contract-invalid rather than masking as clean EOF.

Adverse Event Warning: Beyond Raw AUROC

Balancing Sensitivity, False Alarm Rates, and Decision Windows

Clinical Evaluation Dimensions

- **Event Sensitivity:** Proportion of actual spike episodes detected in advance.
- **False Alerts per Patient-Day (FA/day):** Ensuring alert thresholds do not overwhelm clinical workflows.
- **Effective Lead Time:** Median advance warning window (minutes before glucose breaches threshold).
- **Decision Actionability:** Whether the warning provides sufficient time for dietary adjustment.

Evaluator vs. Serving Policy Disclosures

- **Serving Policy:** Employs stateful hysteresis (Watch/Warning) and 120-min suppression for UI display.
- **Scientific Evaluator:** Uses independent, stateless point-in-time scoring rules for benchmark comparisons.
- **Warning Lead Time Limit:** A 120-min prediction horizon does *not* imply a 120-min warning lead time. Actual lead times depend on event velocity.

Audit Disclosure: The formal EW-01 scientific evaluation harness remains unexecuted due to incomplete governance sign-offs; no speculative superiority claims over clinical baselines are made.

Explaining the Model Without Claiming Causality

Arithmetic Driver Decomposition in the RiskPanel

Arithmetic Driver Decomposition

The RiskPanel visualizes the relative impact of model features on the predicted glucose trajectory:

- **Learned Correction:** Deep residual network delta over the physics prior.
- **Expected Habitual Meal:** Population-level meal timing probability.
- **Logged Meals Absorbing:** Mechanistic gut absorption from patient logs.
- **Return to Baseline:** Physiological clearance and insulin action.

Critical Interpretability Guardrails

- **Internal Decomposition:** Bars represent the mathematical contribution of components to the model's net output (mg/dL impact).
- **Not Causal Biology:** Feature weights do not establish direct biological causality in a living patient.
- **Unannounced Meal Detection:** Disclosed as a heuristic based on UKF innovation residuals, not verified ground truth.

UI Notice: The dashboard includes an explicit disclaimer: *"Contributions explain internal model arithmetic, not biological causality."*

What-If Simulation: Capability & Boundary

Scenario Exploration API and Current Dashboard State

Implemented Backend API Capability

- serve_pr_tft.py::scenarios endpoint supports counterfactual simulation:
  - Simulated meal intake (grams of carbs).
  - Simulated physical activity (immediate uptake approximation).
  - Skipping habitual meals.
- Re-runs UKF and PR-TFT with modified input vectors.
- Returns scenario quantile bands and estimated peak glucose changes.

Current Implementation Boundaries

- **Dashboard UI:** ScenarioEditor.tsx has interactive controls explicitly disabled (disabled = true).
- **Medication Unsupported:** Pharmacological simulations are rejected (SCENARIO_UNSUPPORTED).
- **Approximation Only:** Activity modeling omits delayed insulin sensitivity effects.
- **Not Causal Treatment Effects:** Scenario responses have not been validated against prospective human clinical trials.

Truth in Advertising: We present the What-If capability as an API-level research architecture, not a completed interactive clinical tool.

The Scientific Evidence Ledger

Separating Verified Implementation, Historical Metrics, and Unexecuted Work

Tier 2: Matched Validation Benchmarks

44 PATIENTS (54,281 ELIGIBLE ANCHORS / 53,849–53,920 EVALUABLE)

- **Matched RMSE (mg/dL · 95% Patient-Cluster CI, B=10,000 draws):**
  - **30m:** PR-TFT **12.36** [11.89–12.84] vs. UKF 15.24 [14.30–16.16] vs. Persist 21.64 [20.44–22.79]
    Δ -9.28 vs Persist [-10.05, -8.43], -2.88 vs Twin [-3.37, -2.38]; Rel Red 42.87% / 18.90% (10k draws < 0)
  - **60m:** PR-TFT **19.53** [18.56–20.48] vs. UKF 23.23 [21.76–24.64] vs. Persist 36.05 [33.85–38.13]
    Δ -16.53 vs Persist [-17.77, -15.17], -3.71 vs Twin [-4.27, -3.15]; Rel Red 45.84% / 15.96% (10k draws < 0)
  - **120m:** PR-TFT **25.03** [23.67–26.27] vs. UKF 28.64 [26.67–30.51] vs. Persist 48.36 [45.06–51.46]
    Δ -23.33 vs Persist [-25.31, -21.23], -3.61 vs Twin [-4.44, -2.81]; Rel Red 48.24% / 12.61% (10k draws < 0)
- **Patient Wins (Exact 95% Binomial CI):** 44/44 (100% [92.0–100%]) at 30m & 60m; 120m: 44/44 (100%) vs Persist, 41/44 (93.2% [81.3–98.6%]) vs Twin. *Cohort-limited Bernoulli assumption.*
- **Subgroup Robustness (RMSE60):** Healthy 19.31 (n=15, 18.5k elig / 18.4k eval), Prediabetes 19.87 (n=19, 23.5k elig / 23.3k eval), Type 2 19.18 (n=10, 12.3k elig / 12.2k eval).
- **Resubstitution Diagnostics:** 80% conformal coverage 80.00% (width 37.3 mg/dL); Brier score 0.0703 (raw 0.1650 on 47.6k at-risk anchors). Evaluated on val split; not clinical operating recommendations.
- *Boundary:* Validation cohort used in model selection; reflects simulated development validation, not human generalization.

Tier 1: Verified Implementation

CONFIRMED BY CODE & TESTS

- Observable-input contract enforcement.
- PR-TFT + UKF S3 model architecture.
- Chunk-invariant SSE proxy & recovery hook.
- 37 Jest suites (346 tests) & 111 backend tests passed.

Tier 3: Unexecuted Work

EXCLUDED FROM CLAIMS

- EW-01 scientific evaluation harness remains unexecuted.
- Demo baseline references missing checkpoint; metrics excluded.
- Zero prospective clinical outcome claims.

Evidence Integrity: We do not manufacture benchmark bars or promote historical training-set validation metrics as prospective generalization proof.

Reproducibility & Scientific Limitations

Synthetic Data Realism Gate 3 and Governance Status

Synthetic Data Realism Audit (Gate 3)

Gate 3 Result: FAILED
Discriminator AUROC: **0.9320** (historical, 90% CI: 0.928–0.936) / **0.9290** (duration-corrected, 90% CI: 0.925–0.933) vs. Target: **≤ 0.65**.

- A gradient-boosted discriminator reliably distinguishes synthetic ODE trajectories from mock-real generator records (top tell: jerk_sd, SHAP 0.8878; roc_max_fall).
- Controlled factorial experiments confirm AR(1) drift (SD 8–16 mg/dL) accounts for partial curvature divergence, but multivariate ODE dynamics remain distinguishable. Null control: AUROC 0.507.
- **Hazard Calibration (Resubstitution Diagnostic):** Empirical resubstitution diagnostics on 47,629 at-risk anchors show 14.2% anchor alert rate (sensitivity 63.0%, precision 58.7%) at threshold 0.35, and 5.3% anchor alert rate (sensitivity 29.9%, precision 74.5%) at threshold 0.60. Evaluated thresholds are empirical diagnostics on the validation set, not approved clinical operating points; anchor alert fractions do not represent event-level alerts per day.

Governance & Environmental Constraints

- **EW-01 Evaluation Unexecuted:** Protocol amendments remain pending co-author sign-offs; no experimental outcomes are reported.
- **Missing Baseline Checkpoint:** pr_tft_demo references an absent checkpoint; its baseline numbers are strictly excluded.
- **Clean Environment Verified:** Node 22, Webpack dev server, Python 3.12 venv, and isolated Git worktree verified.

Honest Science: Full scientific transparency requires reporting negative validation gates rather than concealing simulation limitations.

Demonstrated Scope & Research Roadmap

What is Complete, What is Frozen, and What Lies Ahead

Demonstrated & Frozen in this Release

- **Candidate Release Commit:** 67431af93f4e32d68c9df8c57e7ad24da9419de8 passed all independent reviews and test suites.
- **End-to-End Pipeline:** Live synthetic backend → Next.js SSE proxy → browser patient dashboard verified.
- **Framing Resilience:** Chunk-invariant earliest-delimiter parser verified across all split boundaries.
- **Strict Validation:** Comprehensive fail-closed error handling and contract enforcement.

Future Clinical Research Horizons

- **Real Sensor Ingestion:** Hardware connectivity with Dexcom, Abbott FreeStyle Libre, and wearable APIs.
- **Prospective Clinical Trials:** Multi-center observational studies evaluating actual glycemic variability and HbA1c impact.
- **Causal Counterfactuals:** Clinically validated pharmacodynamic and lifestyle intervention models.
- **Interactive UI Scenarios:** Connecting What-If controls after human efficacy verification.

Release Status: Code is verified on candidate commit 67431af9. Full documentation assets, architecture exports, and demo scripts are reconciled.